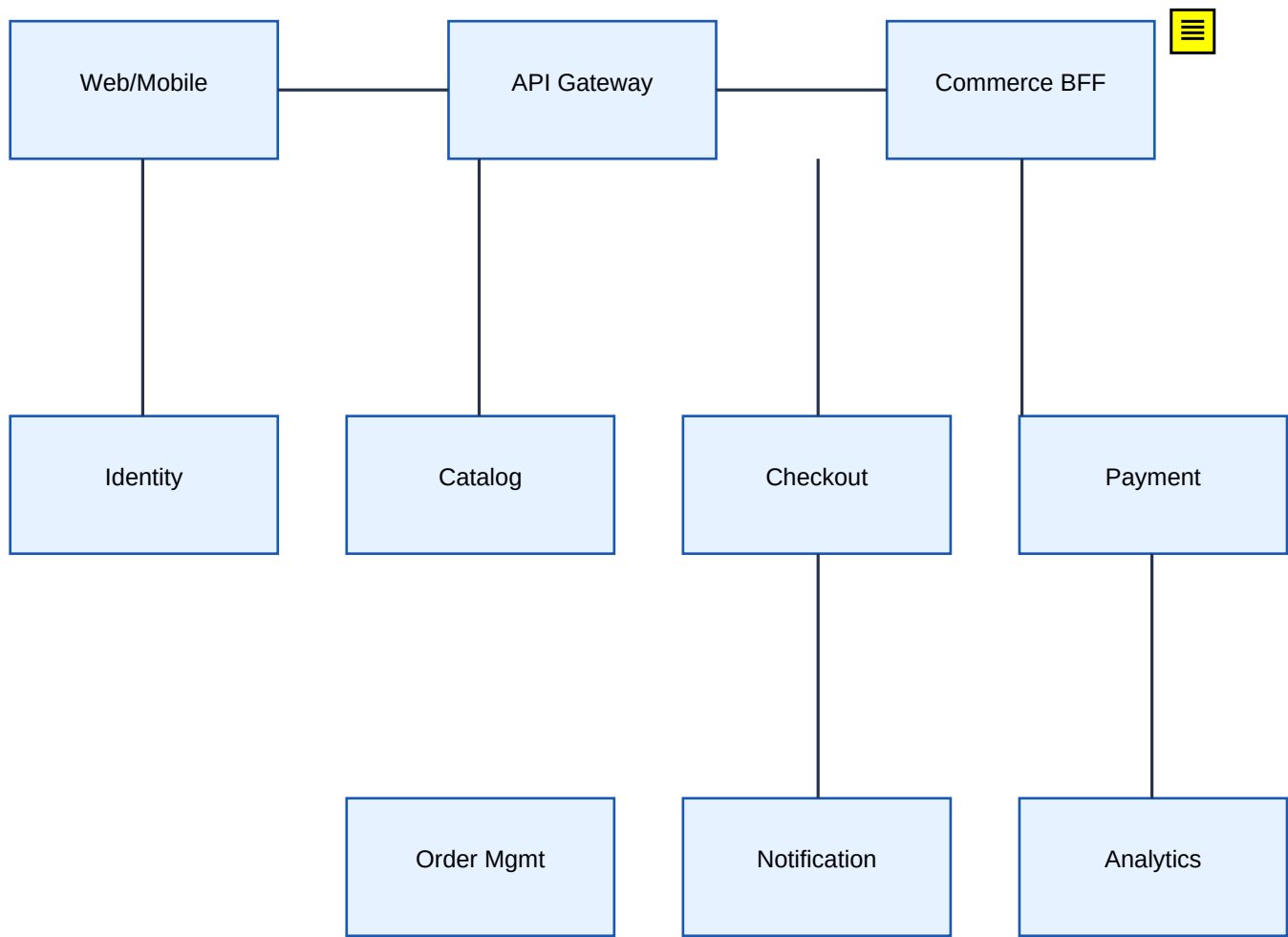


High Level Design

AstraCart Ecommerce Platform

The architecture separates experience, commerce orchestration, identity, catalog, checkout, payment, order management, notification, and observability capabilities. The design supports web and mobile clients through a BFF/API gateway with idempotent command APIs.



Concern	Design Decision	Rationale
Idempotency	Client command id + server idempotency key	Prevents duplicate payment/order creation
Availability	Catalog read path cache + graceful degradation	Home/listing remain usable during recommendation outage
Security	OIDC session + refresh token rotation	Protects account and payment initiation
Observability	Trace id across checkout/payment/order	Supports payment reconciliation triage

Integration Flow: Payment and Order

- 1 Customer confirms checkout quote.
- 2 Checkout service creates payment order with gateway.
- 3 Gateway redirects customer or opens payment intent.
- 4 Redirect returns provisional result to frontend.
- 5 Gateway webhook updates authoritative payment status.
- 6 Order service confirms order, inventory reservation, invoice, and notification.

Event	Producer	Consumer	Failure Handling
PAYMENT_INITIATED	Checkout	Payment	Retry with same quote id
PAYMENT_AUTHORIZED	Gateway	Order	Webhook signature verification
PAYMENT_FAILED	Gateway	Checkout UI	Show retry; retain cart
ORDER_CONFIRMED	Order	Notification	Retry notification without duplicating order
REFUND_REQUESTED	Returns	Payment	Async status visible in order history

